

Age
 ↳ young
 ↳ middle
 ↳ Senior

own-home
 ↳ yes
 ↳ No

Income
 ↳ high
 ↳ medium
 ↳ low

Credit-history

↳ good
 ↳ fair
 ↳ poor

① Age

II	IIII
yes	NO
young	

$$\text{Gini} = 1 - \left[\left(\frac{2}{6} \right)^2 + \left(\frac{4}{6} \right)^2 \right] = 0.4444$$

IIII	
yes	NO
middle	

$$\text{Gini} = 1 - \left[\left(\frac{4}{4} \right)^2 + \left(\frac{0}{4} \right)^2 \right] = 0$$

Age

III
Yes

II
No

Senior

$$\text{Gini} = 1 - \left[\left(\frac{3}{5} \right)^2 + \left(\frac{2}{5} \right)^2 \right] = 0.48$$

$$\text{Age Error} : \frac{6}{15} (0.494) + \frac{4}{15} (0) + \frac{5}{15} (0.48)$$

$$\boxed{\text{Age Error} = 0.3376}$$

Income

III
Yes

I
No

High

$$\text{Gini} = 1 - \left[\left(\frac{3}{4} \right)^2 + \left(\frac{1}{4} \right)^2 \right] = 0.375$$

III
Yes

III
No

Medium

$$\text{Gini} = 1 - \left[\left(\frac{3}{6} \right)^2 + \left(\frac{3}{6} \right)^2 \right] = 0.5$$

III
Yes

II
No

Low

$$\text{Gini} = 1 - \left[\left(\frac{3}{5} \right)^2 + \left(\frac{2}{5} \right)^2 \right] = 0.48$$

Income

$$\text{Error} = \frac{4}{15}(0.375) + \frac{6}{15}(0.5) + \frac{5}{15}(0.48)$$

$$\boxed{\text{Error Income} = 0.46}$$

Own-home

III || |
yes no Gini = $1 - \left[\left(\frac{7}{8} \right)^2 + \left(\frac{1}{8} \right)^2 \right] = 0.219$
Yes

|| III
yes no Gini = $1 - \left[\left(\frac{2}{7} \right)^2 + \left(\frac{5}{7} \right)^2 \right] = 0.408$

NO

own-home Error = $\frac{8}{15}(0.219) + \frac{7}{15}(0.408)$

$$\boxed{\text{own-home Error} = 0.3072}$$

Credit-history

$$\begin{array}{cc} \text{III} & \text{I} \\ \text{yes} & \text{no} \\ \text{good} & \end{array} \quad \text{Gini} = 1 - \left[\left(\frac{5}{6} \right)^2 + \left(\frac{1}{6} \right)^2 \right] = 0.278$$

$$\begin{array}{cc} \text{III} & \text{II} \\ \text{yes} & \text{no} \\ \text{fair} & \end{array} \quad \text{Gini} = 1 - \left[\left(\frac{3}{5} \right)^2 + \left(\frac{2}{5} \right)^2 \right] = 0.48$$

$$\begin{array}{cc} \text{I} & \text{III} \\ \text{yes} & \text{no} \\ \text{poor} & \end{array} \quad \text{Gini} = 1 - \left[\left(\frac{1}{4} \right)^2 + \left(\frac{3}{4} \right)^2 \right] = 0.375$$

$$\text{Error} = \frac{6}{15} (0.278) + \frac{5}{15} (0.48) + \frac{4}{15} (0.375)$$

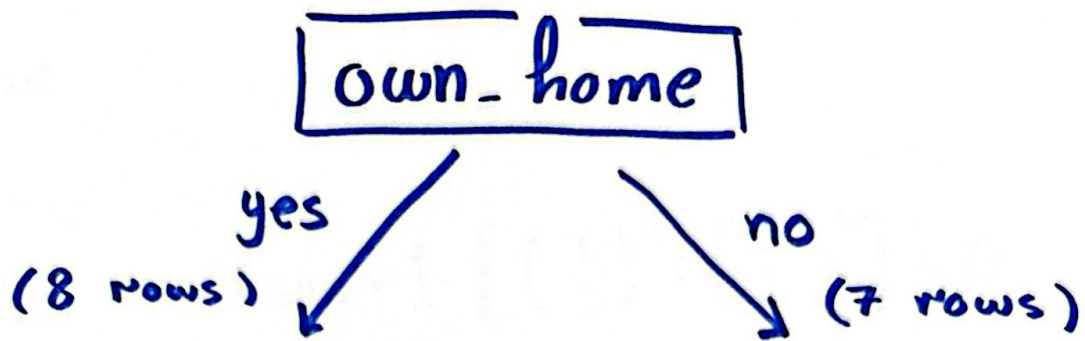
$\text{Credit-history Error} = 0.3712$

Age : 0.3376

Income : 0.4600

Own-home : 0.3072 ★ winner

Credit-history : 0.3712



Target = yes

7 rows

target = no

1 row

Pure ✓

Target = yes

2 rows

Target = no

5 rows

own_home = yes

Age :

II I
yes no
 young

$$\text{Gini} = 1 - \left[\left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right] = 0.444$$

III
yes no
 middle

$$\text{Gini} = 1 - \left[\left(\frac{3}{3} \right)^2 + \left(\frac{0}{3} \right)^2 \right] = 0$$

II
yes no
 Senior

$$\text{Gini} = 1 - \left[\left(\frac{2}{2} \right)^2 + \left(\frac{0}{2} \right)^2 \right] = 0$$

$$\text{Error} = \frac{3}{8} (0.444)$$

$$\boxed{\text{Age Error} = 0.1665}$$

Income

III

yes no

high

$$Gini = 1 - \left[\left(\frac{3}{3} \right)^2 + \left(\frac{0}{7} \right)^2 \right] = 0$$

I
yes

I
no

medium

$$Gini = 1 - \left[\left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2 \right] = 0.5$$

III

yes

no

low

$$Gini = 1 - \left[\left(\frac{3}{3} \right)^2 + \left(\frac{0}{3} \right)^2 \right] = 0$$

$$Error = \frac{2}{8} (0.5)$$

$$\boxed{\text{Income Error} = 0.125}$$

Credit - history

III

yes

no

Good

$$Gini = 1 - \left[\left(\frac{3}{3} \right)^2 + \left(\frac{0}{3} \right)^2 \right] = 0$$

III

yes

no

fair

$$Gini = 1 - \left[\left(\frac{3}{3} \right)^2 + \left(\frac{0}{3} \right)^2 \right] = 0$$

I

I

yes

no

Poor

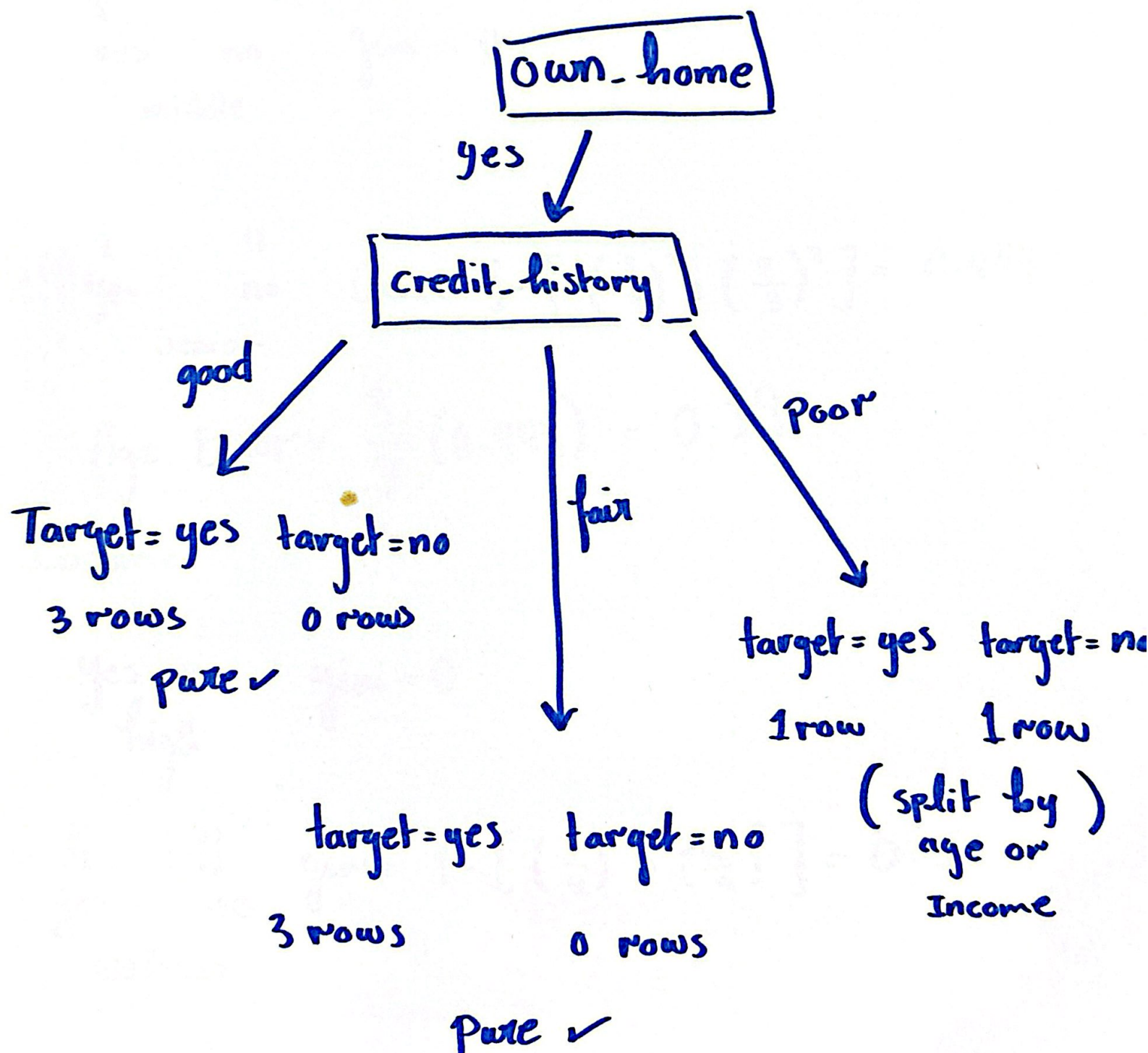
$$Gini = 1 - \left[\left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2 \right] = 0.5$$

$$\text{Credit_history Error} = \frac{2}{8} (0.5) = 0.125$$

Age : 0.1665

Income : 0.125 ★

Credit_history : 0.125 ★ ^{2 winners}
(pick whichever)



OWN home = No

Age

I
yes no
young gini = 0

I
yes no
middle gini = 0

I II
yes no gini = $1 - \left[\left(\frac{1}{3} \right)^2 + \left(\frac{2}{3} \right)^2 \right] = 0.444$
Senior

$$\text{Age error} = \frac{3}{7} (0.444) = 0.19$$

Income

I
yes no
high gini = 0

II II
yes no gini = $1 - \left[\left(\frac{2}{4} \right)^2 + \left(\frac{2}{4} \right)^2 \right] = 0.5$
medium

Income

$$\begin{array}{cc} \text{yes} & \text{no} \\ \text{low} & \end{array} \quad \text{gini} = 1 - \left[\left(\frac{0}{2} \right)^2 + \left(\frac{2}{2} \right)^2 \right] = 0$$

$$\text{Income Error} = \frac{4}{7} (0.5) = 0.286$$

Credit history

$$\begin{array}{cc} \text{yes} & \text{no} \\ \text{good} & \end{array} \quad \text{gini} = 1 - \left[\left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right] = 0.444$$

$$\begin{array}{cc} \text{yes} & \text{no} \\ \text{fair} & \end{array} \quad \text{gini} = 0$$

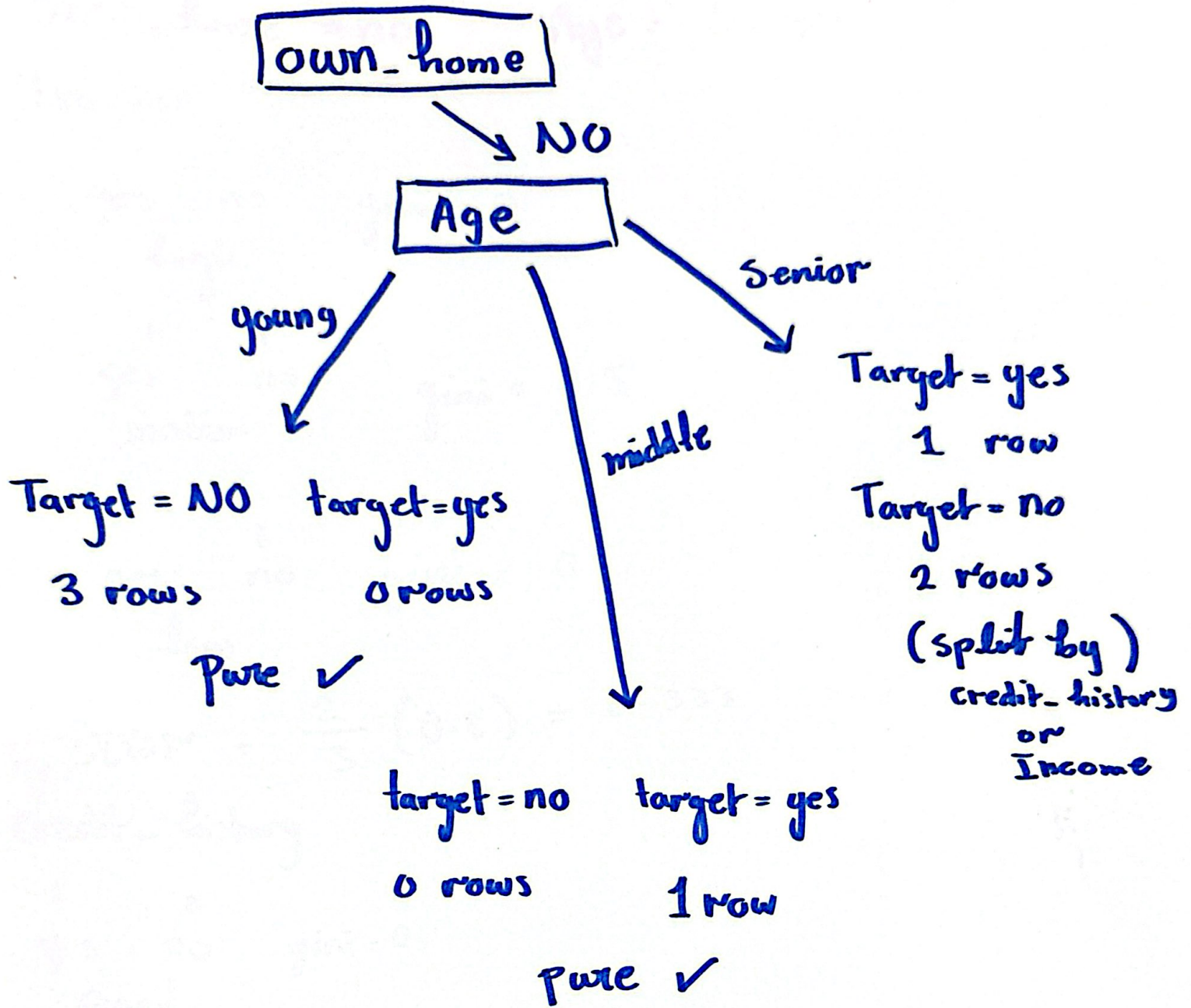
$$\begin{array}{cc} \text{yes} & \text{no} \\ \text{poor} & \end{array} \quad \text{Gini} = 0$$

$$\text{Credit history Error} = \frac{3}{7} (0.444) = 0.19$$

Age: 0.19 *

Income: 0.286 2 winners
(pick any)

Credit history: 0.19 *



own_home = yes

Credit-history = Poor

Age

yes no
young

$gini = 0$

yes no
middle

$gini = 0$

Error = 0

yes no senior $gini = 0$

OWN-home = no

Age = Senior

Income

yes no
high gini = 0

yes no
medium gini = 0.5

yes no
low gini = 0

$$\text{Error} = \frac{2}{3} (0.5) = 0.333$$

Credit-history

yes no
Good gini = 0

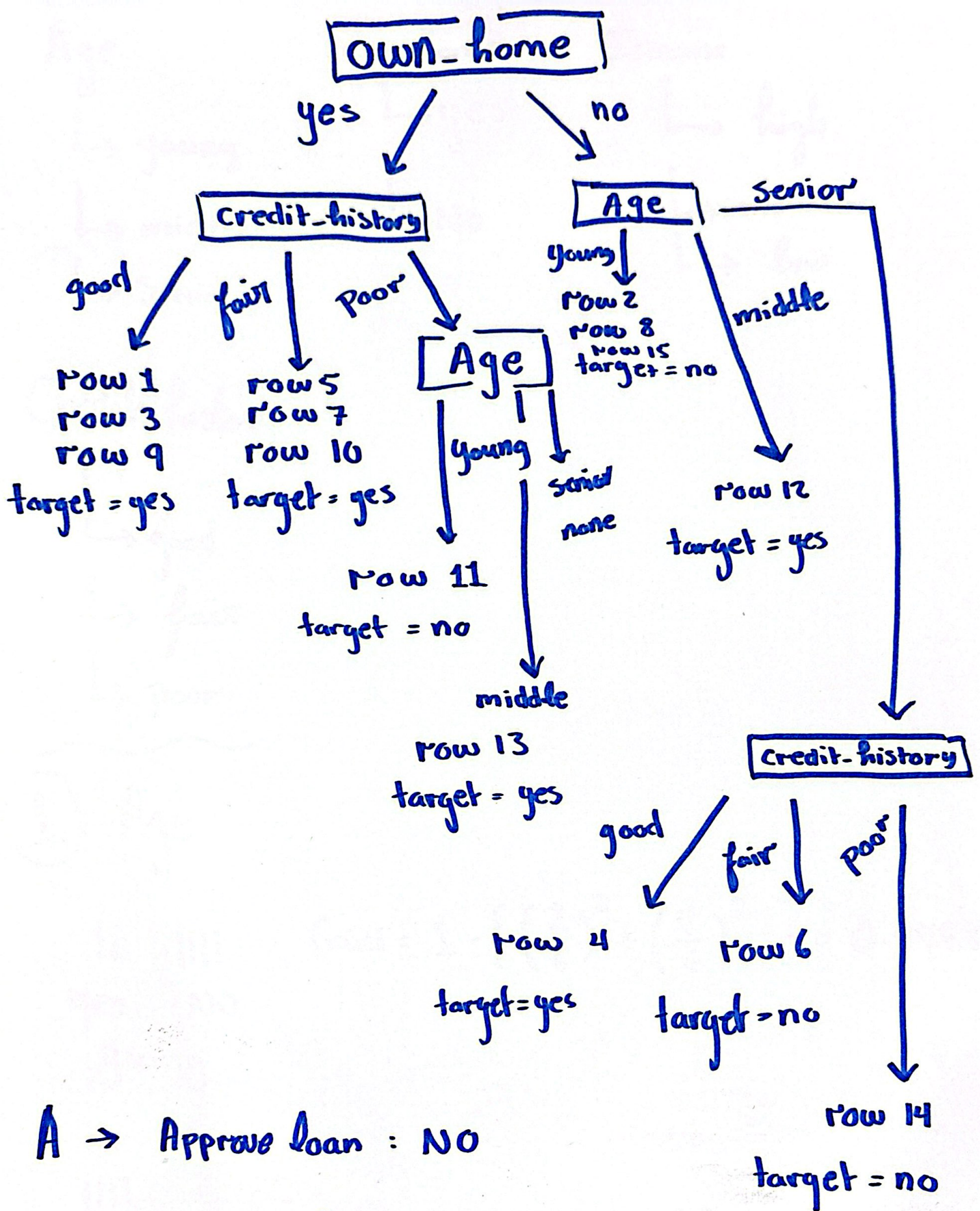
yes no
fair gini = 0

$$\text{Error} = 0$$

yes no
poor gini = 0

Income : 0.333

Credit-history : 0 * winner



A → Approve loan : NO

B → Approve loan : yes